

01_eda

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1 01 — Análise Exploratória de Dados (EDA)

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Base: Building Sites Power Consumption (Möbius, 2021 — Kaggle)

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
import seaborn as sns
from statsmodels.tsa.seasonal import STL
from scipy import stats
import warnings
warnings.filterwarnings('ignore')

# Plot style
plt.rcParams['figure.figsize'] = (14, 5)
plt.rcParams['axes.grid'] = True
plt.rcParams['grid.alpha'] = 0.3
sns.set_palette('tab10')

# Paths
RAW = '../data/raw/'
PROC = '../data/processed/'

print('OK')
```

OK

1.1 1. Coleta de Dados

A base é composta por quatro arquivos CSV relacionados por identificadores de prédio (SiteId).

```
[2]: # Carrega os arquivos brutos
df_train = pd.read_csv(RAW + 'plfec-train-data.csv', sep=';')
df_weather = pd.read_csv(RAW + 'plfec-weather.csv', sep=';')
df_holidays = pd.read_csv(RAW + 'plfec-holidays.csv', sep=';')
df_meta = pd.read_csv(RAW + 'plfec-metadata.csv', sep=';')
```

```

print('Dimensões dos datasets:')
print(f'  train-data : {df_train.shape}')
print(f'  weather    : {df_weather.shape}')
print(f'  holidays    : {df_holidays.shape}')
print(f'  metadata     : {df_meta.shape}')

```

Dimensões dos datasets:

```

train-data : (6559830, 5)
weather    : (3957035, 4)
holidays   : (8387, 3)
metadata    : (267, 11)

```

```

[3]: # Descrição dos campos
print('=== df_train ===')
print(df_train.dtypes)
print(df_train.head(3))
print('\n=== df_weather ===')
print(df_weather.dtypes)
print(df_weather.head(3))
print('\n=== df_holidays ===')
print(df_holidays.dtypes)
print(df_holidays.head(3))
print('\n=== df_meta ===')
print(df_meta.dtypes)
print(df_meta.head(3))

```

=== df_train ===

```

obs_id      int64
SiteId      int64
Timestamp    str
ForecastId  int64
Value       float64

```

dtype: object

	obs_id	SiteId	Timestamp	ForecastId	Value
0	4852050	42	2016-10-18T02:45:00+00:00	1087	26397.049623
1	1638923	42	2016-10-18T11:45:00+00:00	1087	42958.364641
2	5748910	42	2016-10-18T20:45:00+00:00	1087	27096.919666

=== df_weather ===

```

Timestamp    str
Temperature   float64
Distance      float64
SiteId       int64

```

dtype: object

	Timestamp	Temperature	Distance	SiteId
0	2017-03-03T19:00:00+00:00	10.6	27.489346	51
1	2017-03-03T19:20:00+00:00	11.0	28.663082	51

```
2 2017-03-03T20:00:00+00:00          6.3 28.307039          51
```

```
=== df_holidays ===
```

```
Date          str
Holiday        str
SiteId         int64
dtype: object
```

	Date	Holiday	SiteId
0	2016-02-15	Washington's Birthday	1
1	2017-05-29	Memorial Day	1
2	2017-11-23	Thanksgiving Day	1

```
=== df_meta ===
```

```
SiteId          int64
Surface         float64
Sampling         float64
BaseTemperature float64
MondayIsDayOff  bool
TuesdayIsDayOff bool
WednesdayIsDayOff bool
ThursdayIsDayOff bool
FridayIsDayOff  bool
SaturdayIsDayOff bool
SundayIsDayOff  bool
dtype: object
```

	SiteId	Surface	Sampling	BaseTemperature	MondayIsDayOff	\
0	207	7964.873347	30.0	18.0	False	
1	7	15168.125971	30.0	18.0	False	
2	74	424.340663	15.0	18.0	False	

	TuesdayIsDayOff	WednesdayIsDayOff	ThursdayIsDayOff	FridayIsDayOff	\
0	False	False	False	False	
1	False	False	False	False	
2	False	False	False	False	

	SaturdayIsDayOff	SundayIsDayOff
0	True	True
1	True	True
2	True	True

```
[4]: # Converte timestamps
df_train['Timestamp'] = pd.to_datetime(df_train['Timestamp'])
df_weather['Timestamp'] = pd.to_datetime(df_weather['Timestamp'])

print('Período dos dados de energia:')
print(f'  Início : {df_train["Timestamp"].min()}')
print(f'  Fim    : {df_train["Timestamp"].max()}')
```

```
print(f' Sites : {df_train["SiteId"].nunique()} prédios')
```

Período dos dados de energia:

Início : 2009-06-23 06:45:00+00:00

Fim : 2017-12-08 15:30:00+00:00

Sites : 267 prédios

1.2 2. Distribuição de Registros por Prédio

```
[5]: counts = df_train.groupby('SiteId').size().sort_values(ascending=False)

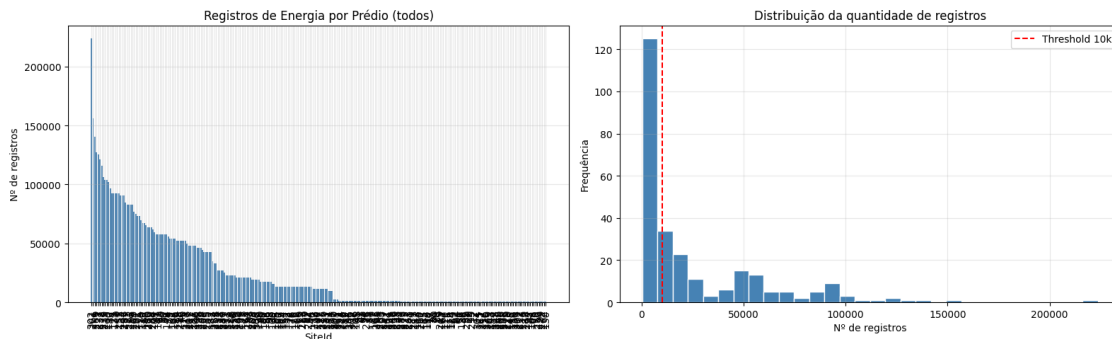
fig, axes = plt.subplots(1, 2, figsize=(16, 5))

# Barras - todos os prédios
axes[0].bar(counts.index.astype(str), counts.values, color='steelblue', width=0.
    ↪8)
axes[0].set_xlabel('SiteId')
axes[0].set_ylabel('Nº de registros')
axes[0].set_title('Registros de Energia por Prédio (todos)')
axes[0].tick_params(axis='x', rotation=90)

# Histograma de distribuição
axes[1].hist(counts.values, bins=30, color='steelblue', edgecolor='white')
axes[1].axvline(10000, color='red', linestyle='--', label='Threshold 10k')
axes[1].set_xlabel('Nº de registros')
axes[1].set_ylabel('Frequência')
axes[1].set_title('Distribuição da quantidade de registros')
axes[1].legend()

plt.tight_layout()
plt.show()

print(f'Prédios com mais de 10.000 registros: {(counts > 10000).sum()}')
print(f'Prédios com mais de 5.000 registros : {(counts > 5000).sum()}')
```



Prédios com mais de 10.000 registros: 139
 Prédios com mais de 5.000 registros : 142

```
[6]: # Seleciona prédios candidatos (>10k registros)
CANDIDATES = counts[counts > 10000].index.tolist()
print(f'Prédios selecionados (>10k registros): {CANDIDATES}')

# Estatísticas básicas por prédio candidato
df_cand = df_train[df_train['SiteId'].isin(CANDIDATES)].copy()
stats_df = df_cand.groupby('SiteId')['Value'].
    .agg(['count', 'mean', 'std', 'min', 'max'])
stats_df.columns = ['N', 'Média (kWh)', 'Desvio Padrão', 'Min', 'Max']
stats_df = stats_df.round(2)
print('\nEstatísticas dos prédios candidatos:')
display(stats_df)
```

Prédios selecionados (>10k registros): [302, 261, 6, 269, 271, 237, 8, 276, 233, 235, 190, 33, 14, 123, 121, 9, 232, 122, 22, 298, 287, 25, 221, 295, 19, 200, 197, 201, 115, 120, 119, 87, 140, 280, 270, 88, 284, 109, 41, 141, 63, 49, 50, 62, 149, 74, 59, 162, 108, 126, 84, 178, 215, 46, 175, 173, 218, 107, 222, 191, 227, 106, 92, 228, 230, 204, 252, 89, 255, 143, 183, 2, 281, 268, 234, 212, 134, 250, 277, 151, 273, 129, 286, 16, 259, 135, 260, 294, 131, 26, 27, 154, 292, 93, 209, 203, 198, 300, 199, 40, 39, 180, 195, 86, 99, 98, 100, 42, 185, 160, 167, 169, 174, 153, 152, 73, 72, 76, 176, 186, 112, 66, 20, 117, 283, 60, 57, 267, 54, 146, 77, 223, 96, 247, 127, 285, 216, 53, 231]

Estatísticas dos prédios candidatos:

	N	Média (kWh)	Desvio Padrão	Min	Max
SiteId					
2	34486	67591.69	54961.98	12558.21	306452.81
6	139716	27682.73	10620.03	13341.72	71063.18
8	114962	37891.42	18922.37	0.00	98200.96
9	91926	3075.19	1885.24	198.54	14484.85
14	91676	32619.19	7890.22	7353.17	62284.36
...	...	...	...	...	...
294	20291	37029.51	25010.03	0.00	157095.37
295	79381	27118.18	18546.26	0.00	125020.55
298	90188	9895.34	19577.64	428.93	4872706.06
300	19136	13067.97	8584.28	6444.48	60331.59
302	223620	44718.28	6735.55	0.00	87921.03

[139 rows x 5 columns]

1.3 3. Análise Temporal do Prédio de Referência (SiteId = 40)

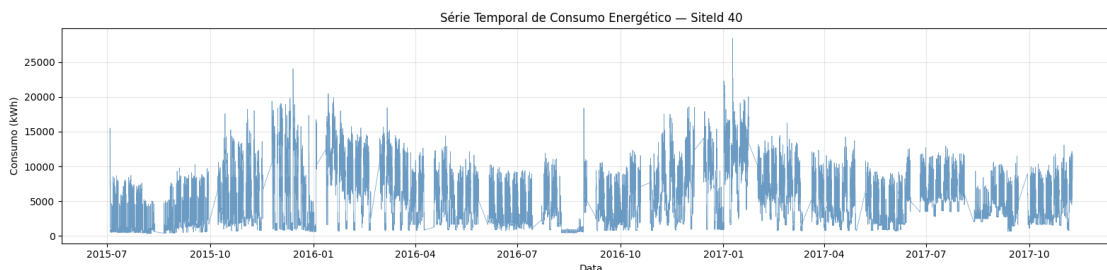
```
[7]: SITE = 40
df40 = df_train[df_train['SiteId'] == SITE].copy().sort_values('Timestamp').
      ↪reset_index(drop=True)

print(f'SiteId {SITE}:')
print(f'  Registros : {len(df40)}')
print(f'  Início    : {df40["Timestamp"].min()}')
print(f'  Fim       : {df40["Timestamp"].max()}')
print(f'  Consumo médio: {df40["Value"].mean():.2f} kWh')
print(f'  Consumo std  : {df40["Value"].std():.2f} kWh')
```

SiteId 40:

Registros : 17352
Início : 2015-07-03 14:30:00+00:00
Fim : 2017-11-08 13:30:00+00:00
Consumo médio: 5476.04 kWh
Consumo std : 3950.01 kWh

```
[8]: # Série temporal completa
fig, ax = plt.subplots(figsize=(16, 4))
ax.plot(df40['Timestamp'], df40['Value'], linewidth=0.5, alpha=0.8,
      ↪color='steelblue')
ax.set_title(f'Série Temporal de Consumo Energético - SiteId {SITE}')
ax.set_xlabel('Data')
ax.set_ylabel('Consumo (kWh)')
plt.tight_layout()
plt.show()
```



```
[9]: # Extrai features de tempo
df40['hour'] = df40['Timestamp'].dt.hour
df40['dayofweek'] = df40['Timestamp'].dt.dayofweek # 0=Seg, 6=Dom
df40['month'] = df40['Timestamp'].dt.month
df40['is_weekend'] = df40['dayofweek'].isin([5, 6]).astype(int)

# Padrão diário
```

```

hourly = df40.groupby('hour')['Value'].mean()

# Padrão semanal
daily = df40.groupby('dayofweek')['Value'].mean()
day_names = ['Seg', 'Ter', 'Qua', 'Qui', 'Sex', 'Sáb', 'Dom']

# Padrão mensal
monthly = df40.groupby('month')['Value'].mean()
month_names = ['Jan', 'Fev', 'Mar', 'Abr', 'Mai', 'Jun', 'Jul', 'Ago', 'Set', 'Out', 'Nov', 'Dez']

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

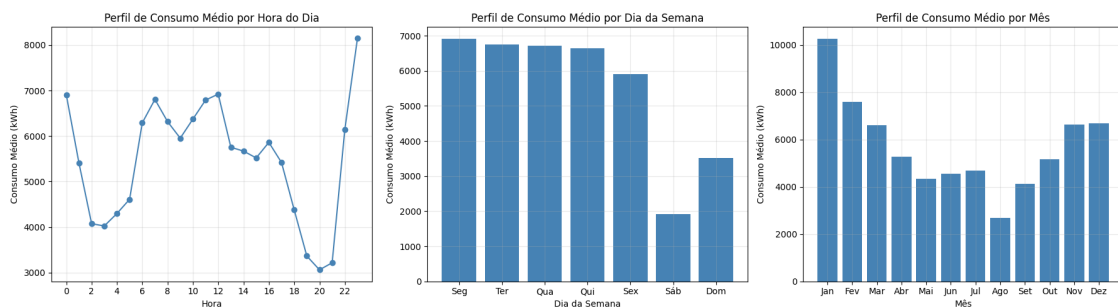
axes[0].plot(hourly.index, hourly.values, marker='o', color='steelblue')
axes[0].set_title('Perfil de Consumo Médio por Hora do Dia')
axes[0].set_xlabel('Hora')
axes[0].set_ylabel('Consumo Médio (kWh)')
axes[0].set_xticks(range(0, 24, 2))

axes[1].bar(range(7), daily.values, color='steelblue', tick_label=day_names)
axes[1].set_title('Perfil de Consumo Médio por Dia da Semana')
axes[1].set_xlabel('Dia da Semana')
axes[1].set_ylabel('Consumo Médio (kWh)')

axes[2].bar(range(1, 13), monthly.values, color='steelblue',
            tick_label=month_names)
axes[2].set_title('Perfil de Consumo Médio por Mês')
axes[2].set_xlabel('Mês')
axes[2].set_ylabel('Consumo Médio (kWh)')

plt.tight_layout()
plt.show()

```

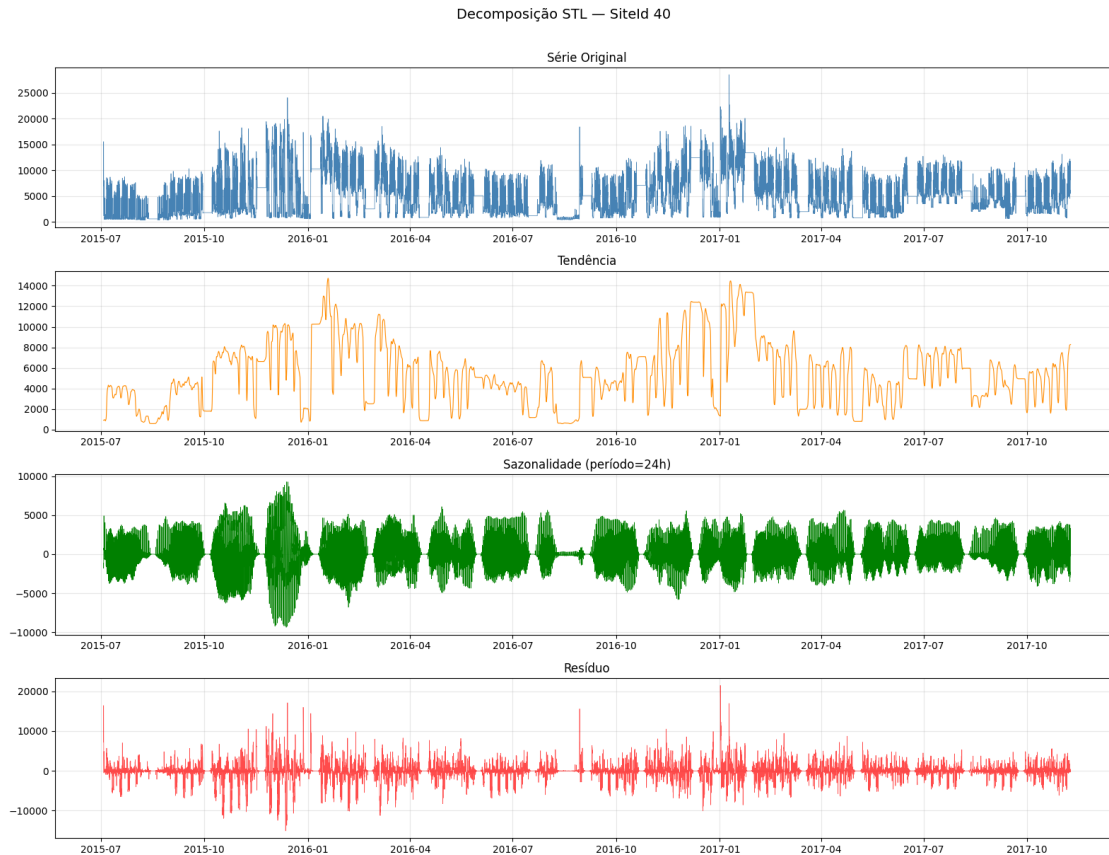


1.4 4. Decomposição STL da Série Temporal

```
[10]: # Resamplea para frequência horária regular (mediana por hora)
df40_hourly = df40.set_index('Timestamp')['Value'].resample('1h').median().
    ↪ffill()

# STL com período de 24h (sazonalidade diária)
stl = STL(df40_hourly, period=24, robust=True)
result = stl.fit()

fig, axes = plt.subplots(4, 1, figsize=(16, 12))
axes[0].plot(df40_hourly.index, df40_hourly.values, linewidth=0.5, ↪
    ↪color='steelblue')
axes[0].set_title('Série Original')
axes[1].plot(result.trend.index, result.trend.values, linewidth=0.8, ↪
    ↪color='darkorange')
axes[1].set_title('Tendência')
axes[2].plot(result.seasonal.index, result.seasonal.values, linewidth=0.5, ↪
    ↪color='green')
axes[2].set_title('Sazonalidade (período=24h)')
axes[3].plot(result.resid.index, result.resid.values, linewidth=0.5, ↪
    ↪color='red', alpha=0.7)
axes[3].set_title('Resíduo')
for ax in axes:
    ax.set_xlabel('')
    ax.grid(True, alpha=0.3)
plt.suptitle(f'Decomposição STL - SiteId {SITE}', fontsize=14, y=1.01)
plt.tight_layout()
plt.show()
```

1.5 5. Correlação de Pearson entre Variáveis

```
[11]: # Cruza energia com temperatura para SiteId 40
# Usa a menor distância ao prédio (mesma abordagem do TCC2)
df_w40 = df_weather[df_weather['SiteId'] == SITE].copy()
df_w40['Timestamp'] = pd.to_datetime(df_w40['Timestamp'])
df_w40 = df_w40.sort_values(['Timestamp', 'Distance']).
↳ drop_duplicates(subset='Timestamp', keep='first')

df_merged = pd.merge(df40, df_w40[['Timestamp', 'Temperature']],
↳ on='Timestamp', how='inner')
df_merged['hour_num'] = df_merged['hour']

# Adiciona feriados
df_hol = df_holidays[df_holidays['SiteId'] == SITE].copy()
df_hol['Date'] = pd.to_datetime(df_hol['Date']).dt.date
df_merged['date'] = df_merged['Timestamp'].dt.date
df_merged['is_holiday'] = df_merged['date'].isin(df_hol['Date']).astype(int)
```

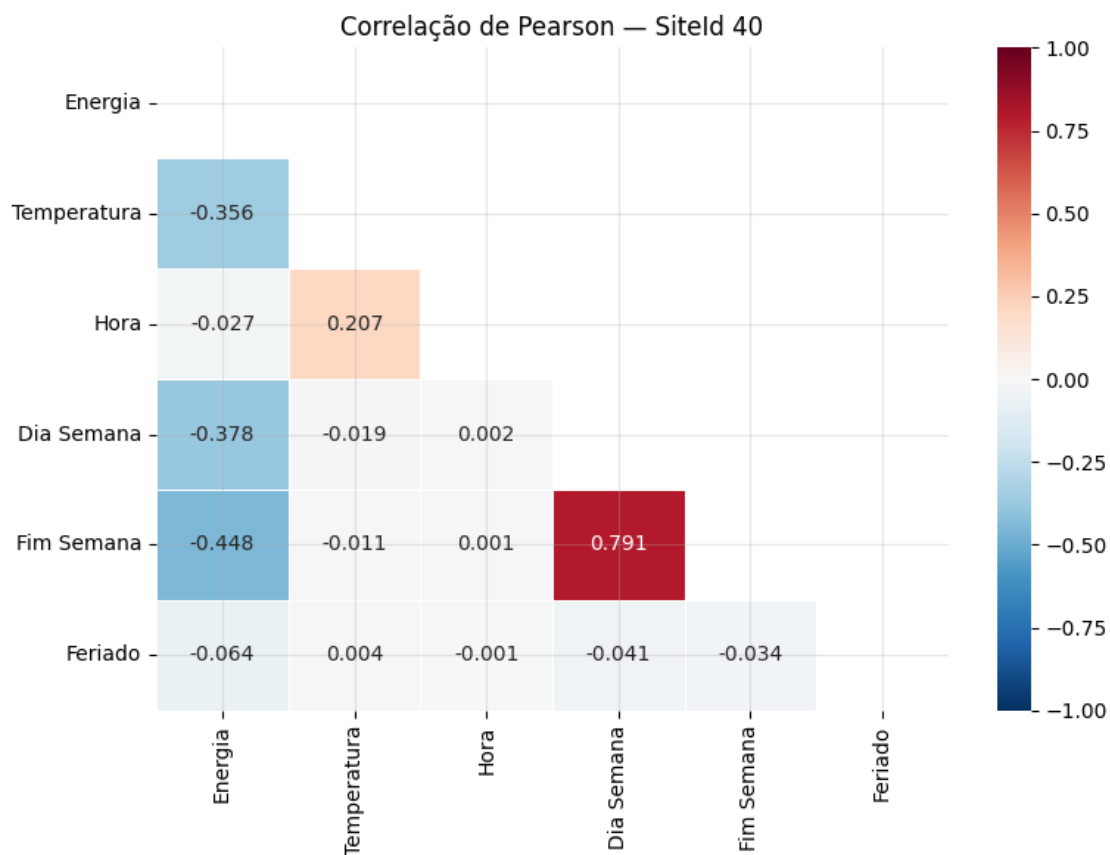
```

corr_cols = ['Value', 'Temperature', 'hour_num', 'dayofweek', 'is_weekend',
             ↪ 'is_holiday']
corr_labels = ['Energia', 'Temperatura', 'Hora', 'Dia Semana', 'Fim Semana',
             ↪ 'Feriado']
corr_matrix = df_merged[corr_cols].corr()
corr_matrix.index = corr_labels
corr_matrix.columns = corr_labels

fig, ax = plt.subplots(figsize=(8, 6))
mask = np.triu(np.ones_like(corr_matrix, dtype=bool))
sns.heatmap(corr_matrix, annot=True, fmt='.3f', cmap='RdBu_r', center=0,
            vmin=-1, vmax=1, ax=ax, mask=mask, linewidths=0.5)
ax.set_title(f'Correlação de Pearson - SiteId {SITE}')
plt.tight_layout()
plt.show()

print('Correlações com Energia:')
print(corr_matrix['Energia'].drop('Energia').sort_values(key=abs,
             ↪ ascending=False).to_string())

```



Correlações com Energia:

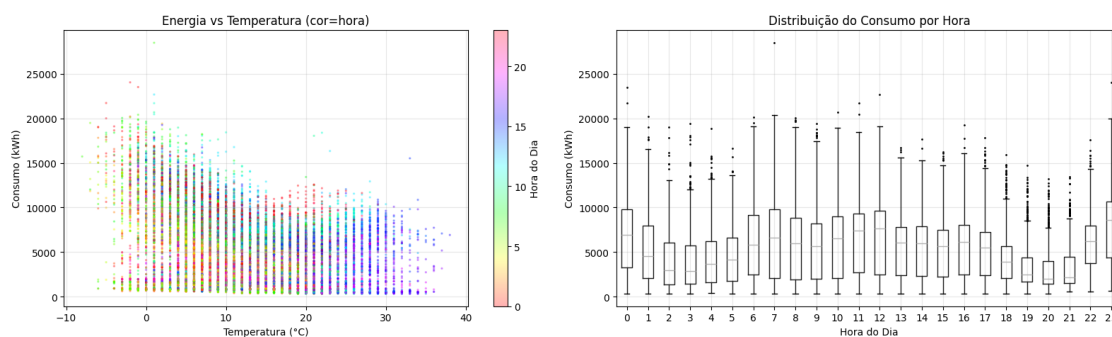
Fim Semana -0.447600
Dia Semana -0.377915
Temperatura -0.356072
Feriado -0.064320
Hora -0.027284

```
[12]: # Scatter: Energia vs Temperatura colorido por hora
fig, axes = plt.subplots(1, 2, figsize=(16, 5))

sc = axes[0].scatter(df_merged['Temperature'], df_merged['Value'],
                    c=df_merged['hour'], cmap='hsv', alpha=0.3, s=2)
plt.colorbar(sc, ax=axes[0], label='Hora do Dia')
axes[0].set_xlabel('Temperatura (°C)')
axes[0].set_ylabel('Consumo (kWh)')
axes[0].set_title('Energia vs Temperatura (cor=hora)')

# Boxplot por hora
df_merged.boxplot(column='Value', by='hour', ax=axes[1],
                  flierprops=dict(markersize=1))
axes[1].set_title('Distribuição do Consumo por Hora')
axes[1].set_xlabel('Hora do Dia')
axes[1].set_ylabel('Consumo (kWh)')
plt.suptitle('')

plt.tight_layout()
plt.show()
```



1.6 6. Detecção de Outliers

```
[13]: # IQR
Q1 = df40['Value'].quantile(0.25)
Q3 = df40['Value'].quantile(0.75)
IQR = Q3 - Q1
lower_iqr = Q1 - 1.5 * IQR
```

```

upper_iqr = Q3 + 1.5 * IQR
outliers_iqr = df40[(df40['Value'] < lower_iqr) | (df40['Value'] > upper_iqr)]

# Z-score
df40['z_score'] = np.abs(stats.zscore(df40['Value']))
outliers_z = df40[df40['z_score'] > 3]

print(f'Outliers por IQR : {len(outliers_iqr)} ({100*len(outliers_iqr)/
↳len(df40):.1f}%)')
print(f'Outliers por Z>3 : {len(outliers_z)} ({100*len(outliers_z)/len(df40):.
↳1f}%)')

fig, axes = plt.subplots(1, 2, figsize=(16, 5))

axes[0].plot(df40['Timestamp'], df40['Value'], linewidth=0.4,
↳color='steelblue', alpha=0.7, label='Normal')
axes[0].scatter(outliers_iqr['Timestamp'], outliers_iqr['Value'],
↳color='red', s=10, zorder=5, label='Outlier (IQR)')
axes[0].axhline(upper_iqr, color='orange', linestyle='--', linewidth=1,
↳label=f'Limite sup IQR ({upper_iqr:.0f})')
axes[0].axhline(lower_iqr, color='orange', linestyle='--', linewidth=1,
↳label=f'Limite inf IQR ({lower_iqr:.0f})')
axes[0].set_title('Outliers por IQR')
axes[0].set_xlabel('Data')
axes[0].set_ylabel('Consumo (kWh)')
axes[0].legend(fontsize=8)

axes[1].hist(df40['Value'], bins=60, color='steelblue', edgecolor='white',
↳density=True)
axes[1].axvline(upper_iqr, color='orange', linestyle='--', label='Limite IQR')
axes[1].axvline(df40['Value'].mean() + 3*df40['Value'].std(), color='red',
↳linestyle='--', label='Média+3 ')
axes[1].set_title('Distribuição do Consumo')
axes[1].set_xlabel('Consumo (kWh)')
axes[1].set_ylabel('Densidade')
axes[1].legend()

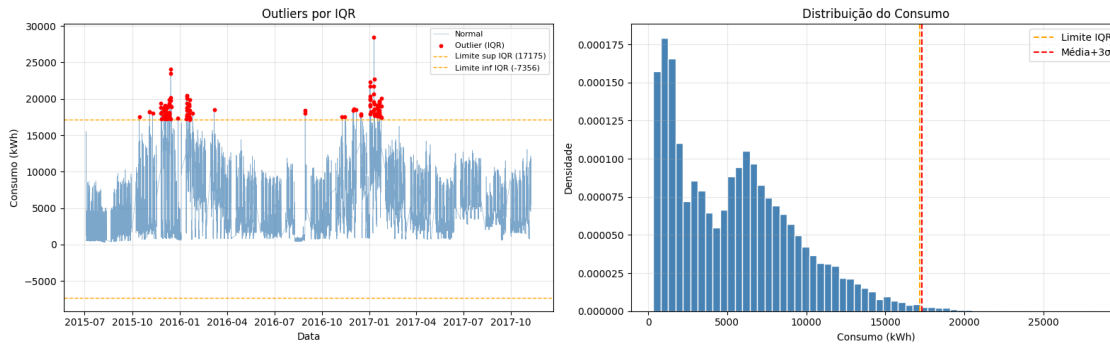
plt.tight_layout()
plt.show()

```

```

Outliers por IQR : 116 (0.7%)
Outliers por Z>3 : 0 (0.0%)

```



```
[14]: # Salva o dataframe mesclado (base para notebooks seguintes)
df_merged_clean = df_merged.drop(columns=['z_score', 'date'], errors='ignore')
df_merged_clean.to_csv(PROC + 'df_site40_merged.csv', index=False)
print(f'Salvo: {PROC}df_site40_merged.csv - shape: {df_merged_clean.shape}')
df_merged_clean.head(3)
```

Salvo: ../data/processed/df_site40_merged.csv - shape: (16995, 12)

```
[14]:
```

	obs_id	SiteId	Timestamp	ForecastId	Value	hour	\
0	5338068	40	2015-07-03 14:30:00+00:00	1002	6997.984367	14	
1	3210822	40	2015-07-03 15:30:00+00:00	1002	15518.012898	15	
2	412101	40	2015-07-03 16:30:00+00:00	1002	6674.491991	16	

	dayofweek	month	is_weekend	Temperature	hour_num	is_holiday
0	4	7	0	31.0	14	0
1	4	7	0	33.0	15	0
2	4	7	0	34.0	16	0